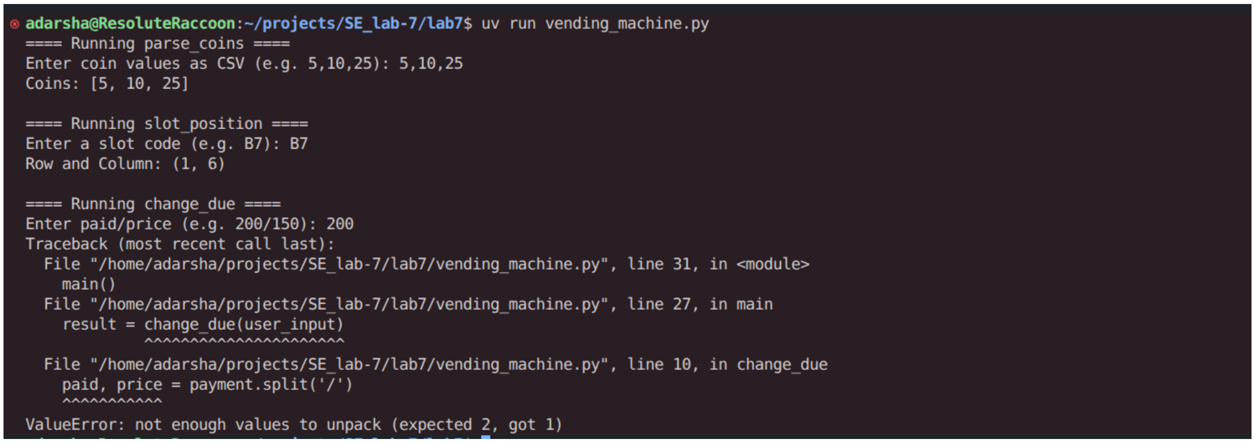
SE LAB-7 Fuzz Testing

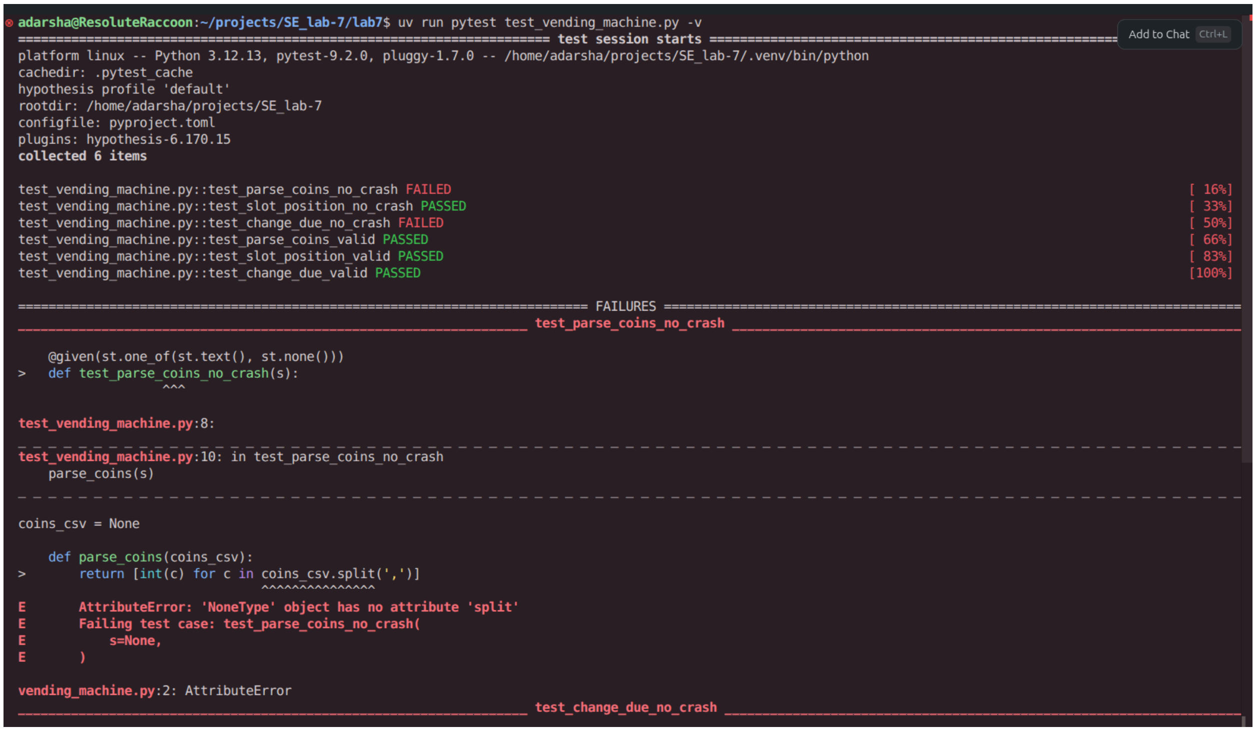
Name: Adarsha E SRN: PES1UG24AM334 Section: AIML ‘F’

1. Run the buggy application:



```
⊗ adarsha@ResoluteRaccoon:~/projects/SE_lab-7/lab7$ uv run vending_machine.py
==== Running parse_coins ====
Enter coin values as CSV (e.g. 5,10,25): 5,10,25
Coins: [5, 10, 25]

==== Running slot_position ====
Enter a slot code (e.g. B7): B7
Row and Column: (1, 6)

==== Running change_due ====
Enter paid/price (e.g. 200/150): 200
Traceback (most recent call last):
  File "/home/adarsha/projects/SE_lab-7/lab7/vending_machine.py", line 31, in <module>
    main()
  File "/home/adarsha/projects/SE_lab-7/lab7/vending_machine.py", line 27, in main
    result = change_due(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^

  File "/home/adarsha/projects/SE_lab-7/lab7/vending_machine.py", line 10, in change_due
    paid, price = payment.split('/')
    ^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

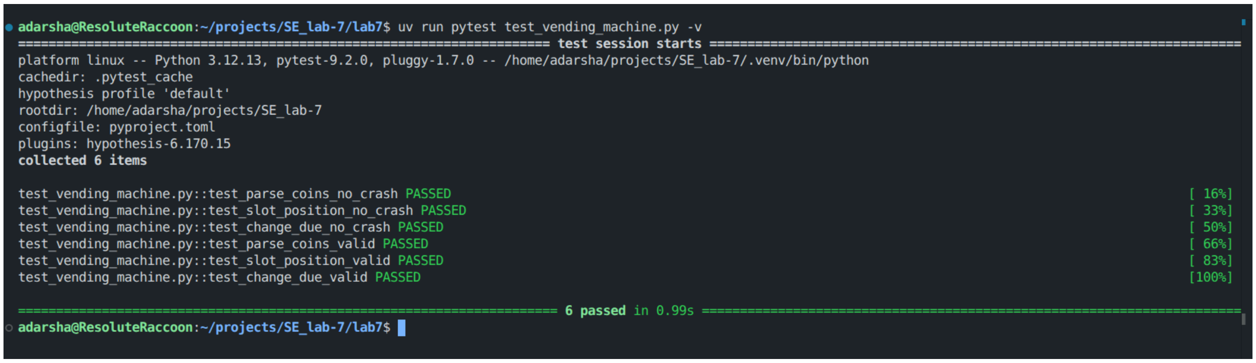
2. Run fuzz tests on original code:



Add to Chat Ctrl+L

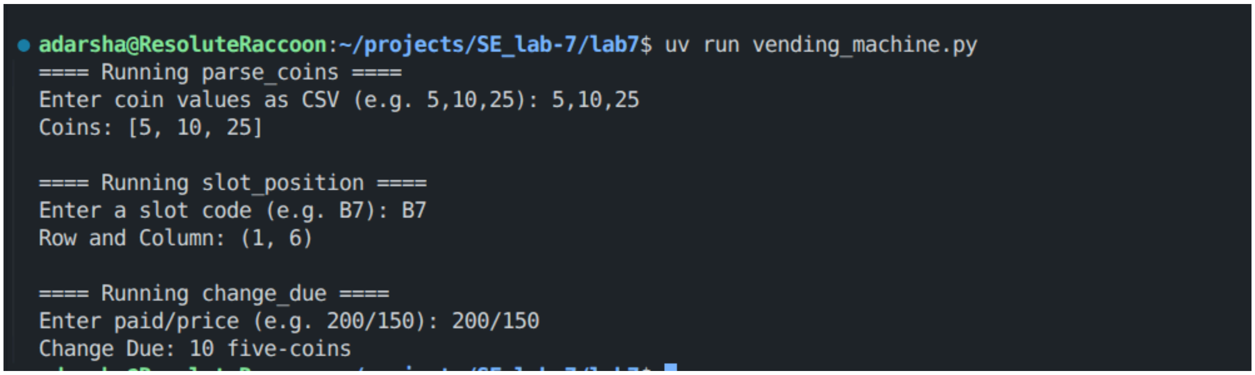
```
⊗ adarsha@ResoluteRaccoon:~/projects/SE_lab-7/lab7$ uv run pytest test_vending_machine.py -v
====================================================================== test session starts ======================================================================
platform linux -- Python 3.12.13, pytest-9.2.0, pluggy-1.7.0 -- /home/adarsha/projects/SE_lab-7/.venv/bin/python
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /home/adarsha/projects/SE_lab-7
configfile: pyproject.toml
plugins: hypothesis-6.170.15
collected 6 items

test_vending_machine.py::test_parse_coins_no_crash FAILED                                                                                          [ 16%]
test_vending_machine.py::test_slot_position_no_crash PASSED                                                                                        [ 33%]
test_vending_machine.py::test_change_due_no_crash FAILED                                                                                           [ 50%]
test_vending_machine.py::test_parse_coins_valid PASSED                                                                                             [ 66%]
test_vending_machine.py::test_slot_position_valid PASSED                                                                                           [ 83%]
test_vending_machine.py::test_change_due_valid PASSED                                                                                              [100%]

=========================================================================== FAILURES ===========================================================================
__________________________________________________________________ test_parse_coins_no_crash __________________________________________________________________

    @given(st.one_of(st.text(), st.none()))
>   def test_parse_coins_no_crash(s):
                   ^^^

test_vending_machine.py:8:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
test_vending_machine.py:10: in test_parse_coins_no_crash
    parse_coins(s)
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

coins_csv = None

    def parse_coins(coins_csv):
>       return [int(c) for c in coins_csv.split(',')]
                                ^^^^^^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'split'
E       Failing test case: test_parse_coins_no_crash(
E           s=None,
E       )

vending_machine.py:2: AttributeError
__________________________________________________________________ test_change_due_no_crash __________________________________________________________________
```

3. Run fuzz tests on corrected code:



```
adarsha@ResoluteRaccoon:~/projects/SE_lab-7/lab7$ uv run pytest test_vending_machine.py -v
====================================================================== test session starts ======================================================================
platform linux -- Python 3.12.13, pytest-9.2.0, pluggy-1.7.0 -- /home/adarsha/projects/SE_lab-7/.venv/bin/python
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /home/adarsha/projects/SE_lab-7
configfile: pyproject.toml
plugins: hypothesis-6.170.15
collected 6 items

test_vending_machine.py::test_parse_coins_no_crash PASSED                                                                                                 [ 16%]
test_vending_machine.py::test_slot_position_no_crash PASSED                                                                                               [ 33%]
test_vending_machine.py::test_change_due_no_crash PASSED                                                                                                  [ 50%]
test_vending_machine.py::test_parse_coins_valid PASSED                                                                                                    [ 66%]
test_vending_machine.py::test_slot_position_valid PASSED                                                                                                  [ 83%]
test_vending_machine.py::test_change_due_valid PASSED                                                                                                     [100%]

======================================================================= 6 passed in 0.99s =======================================================================
adarsha@ResoluteRaccoon:~/projects/SE_lab-7/lab7$
```

4. Run the fixed application:



```
adarsha@ResoluteRaccoon:~/projects/SE_lab-7/lab7$ uv run vending_machine.py
==== Running parse_coins ====
Enter coin values as CSV (e.g. 5,10,25): 5,10,25
Coins: [5, 10, 25]

==== Running slot_position ====
Enter a slot code (e.g. B7): B7
Row and Column: (1, 6)

==== Running change_due ====
Enter paid/price (e.g. 200/150): 200/150
Change Due: 10 five-coins
```